

OSPF Lab Report: Serial Link Configuration Update

1. Serial Link Configuration (R1 to R2)

As requested, R1 is configured as the DCE device with a clock rate of 128000. This is crucial for synchronizing the serial interface.

```
! Configuration on R1 (DCE)
interface Serial0/0/0
 ip address 192.168.12.1 255.255.255.252
 clock rate 128000
 no shutdown

! Configuration on R2 (DTE)
interface Serial0/0/0
 ip address 192.168.12.2 255.255.255.252
 no shutdown
```

Note: The `clock rate` command is only required on the DCE side. The DTE interface automatically detects the clocking provided by the DCE.

2. OSPF Configuration for Serial Links

To ensure optimal OSPF operation over point-to-point links:

```
router ospf 1
 network 192.168.12.0 0.0.0.3 area 0
```

3. Verification

Use the following commands to confirm the serial link is operational:

- `show controllers serial 0/0/0`: Confirms the interface is a DCE and displays the configured clock rate.
- `show ip interface brief`: Checks if the interface is "up/up".

- `show ip ospf neighbor`: Confirms the OSPF adjacency is in the "FULL" state.